

6 Portfolio Metrics

Portfolios are sets of manuscripts that belong to entities like individual researchers, labs, institutions, geographic regions, time periods, academic fields, etc. Grouping manuscripts into these meaningful sets allows Liberata to produce insightful portfolio level metrics.

6.1 Academic Capital

Recall from section 2.1 that academic capital is defined as the sum of the product of shares and weighted citations over some set of papers (??).

$$\mathbf{x}_{M,C} = \sum_{m \in M, c \in C} s_{m,c} \cdot w_m$$

By picking shares belong to different sets of M and C , we can measure the following, which are meant to be illustrative and not exhaustive.

- Set M to be the papers an individual $C_i = a_i \cup p_i \cup r_i$ has shares on. $\mathbf{x}$ in this case represents the individual's career academic contributions and can be a higher resolution, more accurate metric compared to citation count, H-index, and its derivatives. If C is restricted to just the author node, peer reviewer node, or replicator node of this individual, and one can see the individual's relative contributions in original research, academic review, or replication. Restrict M by papers belonging to academic disciplines, and one can give an accurate breakdown of contributions of the individual to different fields of science.
- Set M to be the papers that a lab or institution is involved in and C to be the three roles of all members of that institution. $\mathbf{x}$ in this case measures the total contributions of those institutions to academic research, and this quantity can be further broken down by particular labs, by function (authorship, review, replication), and by academic discipline. One could also measure average contribution or collaborativeness of labs or institutions by examining ratios of $\mathbf{x}_{M,C}$ with the $C_{institution}$ contributors to the $\mathbf{x}_M$ for all contributors (i.e. setting $s_{m,c}$ to 1).
- Set M to be the papers for a particular academic discipline, and one can calculate the relative impacts and sizes of different academic fields.
- Set M to be papers published within a particular time period, and one can calculate and compare the relative impacts and productivity of different eras within academia.
- Set C to be academics from particular geographic regions, and one can calculate the academic output of different parts of the globe, and also further subdivide that by filtering M by in-region and out-region works to see where academic capital is generated and where it flows.

In addition to the above, one can imagine mixing intersections of the conditions mentioned above (individuals, institutions, geographic regions, time periods, academic field), and more, to make custom "portfolios" of academic contributions. Formally, a portfolio is a collection of shares on a set of manuscripts $M' \subseteq M$ owned by a set of contributors $C' \subseteq C$.

$$\Pi_{M',C'} = \{(m, c) \in M' \times C' \mid s(m, c) > 0\} \quad (32)$$

6.2 Academic Taxonomy

The classification of all manuscripts (and portfolios) by academic field implies a taxonomic system to categorize all of academia. Liberata inherits and extends the system of 4 levels of tags from OpenAlex Priem et al. [2022], which is called the 4D system, analogous to the GICS system for classifying companies into industries. For the purposes of this paper, the two can be thought of as functionally identical and a sufficiently accurate and detailed classification system for all analysis purposes. The Liberata taxonomy consists of four levels of granularity.

Liberata's tagging system is of a four layer tree structure, where $d_4 \in D_4$ tags are child nodes of $d_3 \in D_3$ tags which are in turn child nodes of $d_2 \in D_2$ which are finally child nodes of $d_1 \in D_1$ tags. No child node can have multiple parent nodes. All manuscripts in Liberata are labeled with one d_4 tag upon uploading, and automatically inherit d_3, d_2, d_1 parent tags. "Academic field" will be used in this paper to refer to the set of taxonomic tags associated with a manuscript or person.

Table 1: Names, descriptions, and populations of each taxonomic level of Liberata

Entity	Scope / Granularity	Set Size
Domain: D_1	Analogous to schools or faculties	$\sim 10^0$
Department: D_2	Analogous to institutional departments	$\sim 10^1$
Discipline: D_3	Analogous to academic disciplines (majors/minors)	$\sim 10^2$
Direction: D_4	Analogous to research topics/directions	$\sim 10^3$

Two examples of the 4D taxonomy are shown in fig. 2.

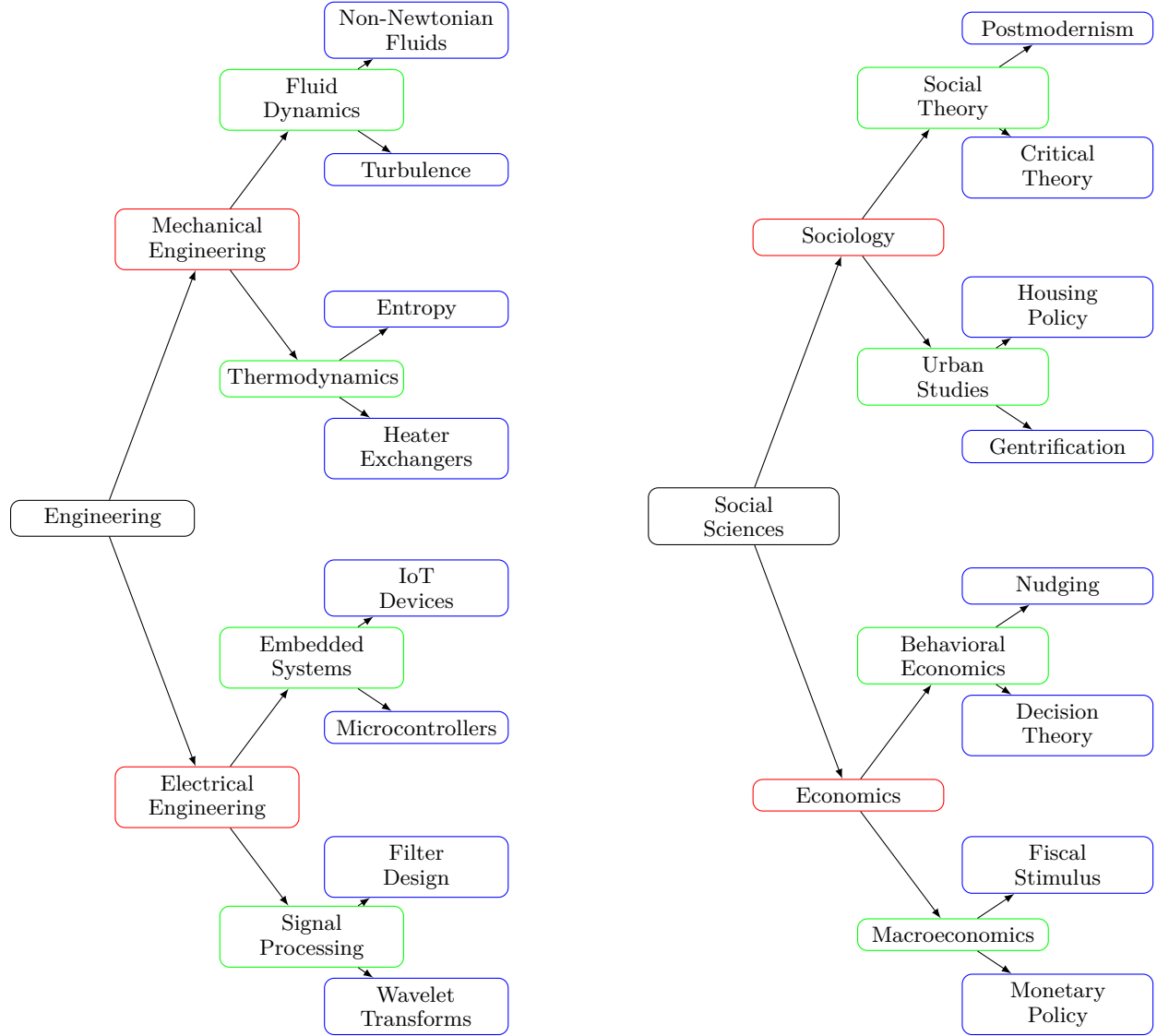


Figure 2: Example tag tree structure illustrating domain, department, discipline, and direction levels in the 4D system.

6.3 Portfolio Mix

On the Liberata platform, in addition to the 4D tags identifying relevant academic fields, additional tags indicating institution, authors, geographic region, and publication year of each manuscript are used. These tags allow for categorization of different works, and choosing combinations of these tags allows for the

construction of an immense range of academic portfolios for further analysis. The first simple analysis is portfolio mix. There are three possible ways to divide an academic portfolio.

Firstly, portfolios can be broken down by academic field to varying levels of granularity according to the 4D system in section 6.2. This is roughly analogous to the multilevel breakdown of equity assets (sector, industry group, industry, sub-industry) by the Global Industry Classification Standard (GICS) widely used by companies such as the S&P and MSCI. While multiple tags are allowed for each level to facilitate literature searching and browsing, authors will indicate upon uploading which tag at each level is the most relevant to the work, and any academic capital the paper accrues will go to those four tags to prevent multi-counting.

Secondly, portfolios can be broken down by contributor role. Shares held by any contributor are labeled by whether that contributor is an original author, a peer reviewer, or a replicator of the academic work.

Thirdly, portfolios can be broken down by time period. This breakdown gives a sense of career contributions across time or institution productivity and competitiveness over its existence.

These three breakdowns, field, role, and time, combined with all the tags that can be used to construct portfolios, allow for widely ranging (albeit finite since each set of tags is finite) quantities to be checked and compared.

6.4 Relevancy & Similarity

Given objects categorized by the 4D tag system, a natural way to measure relevance or similarity of works arises. By choosing a tag level of granularity, i.e. d_4 , one can assemble vectors representing each manuscript by having a 0 for d_4 tags that are not tagged onto the manuscript, and a 1 for d_4 tags that are tagged onto the manuscript. This can be thought of as a taxonomy vector for that manuscript. By computing the cosine similarity between two taxonomy vectors, one can measure the similarity of the two manuscripts from a value of 0 meaning the two works have no shared tags in common, to a value of 1 meaning the two works have exactly the same set of tags in common.

The choice of granularity by level of tag allows for more fine grained or coarse grained comparisons of manuscripts. The cosine similarity measure can also be extended to portfolios of works by having a taxonomy vector for a portfolio being the weighted average (by academic capital or by shares held) of manuscript taxonomy vectors, allowing for the comparison of similarity between portfolios.

Another method to measure relevancy between two manuscripts m_i and m_j , and a way by which to automate the assignment of tags to new works, is to take the rows i and j on the references graph $\mathbf{G}_W$ as co-citation vectors, and compute the cosine similarity between those. If the cosine similarity is found to be close to 1 by some arbitrary threshold, we can assign m_j the same set of tags as m_i . A more complicated analysis can also be done looking at the tags of the referenced works of m_j , (if m_j is the taxonomically unknown work), and computing a weighted average of all of the manuscripts in m_j 's references section and interpreting the resulting value in each component of the vector as the confidence of that tag being appropriate for m_j . This can be then presented to m_j 's author(s) for verification, or just automatically assigned for backward compatibility or if m_j 's authors are unreachable.

The notion of relevancy via cosine similarity, also allows us to construct a relevancy matrix, for search and recommendation purposes. Let $\mathbf{G}_D$ be the relevancy matrix. Then the elements of $\mathbf{G}_D$, g_{ij} are defined as the cosine similarity of the taxonomy vector $\mathbf{d}_i$ of m_i and the taxonomy vector $\mathbf{d}_j$ of m_j :

$$g_{ij} = \frac{\mathbf{d}_i \cdot \mathbf{d}_j}{\|\mathbf{d}_i\| \|\mathbf{d}_j\|} \quad (33)$$

Note that by construction, this matrix is symmetric about the main diagonal, and the values along the main diagonal are 1's. By using fetch vectors to retrieve a row j , and sorting the indices by relevance, we can find the most relevant works to recommend to the author should they want relevant reading. This measure of relevance can be swapped out for a co-citation version of relevance, instead of the tag based measure of relevance.

6.5 Expected Returns

Like with financial assets, the most commonly used portfolios would be that for an individual or institution, although everything mentioned in this subsection does have an analogy to all other possible portfolios or cross sections of manuscripts and their academic capital.

Given a portfolio Π , one can estimate the expected return or growth in academic capital over time, due to different natural publication rates in different academic fields. In addition, one can also estimate the risk or standard deviation from that mean return. This leads to a relatively simple set of metrics that can help institutions or government bodies gain insight into which academic fields are more impactful per unit risk, informing investment choices when resources are limited.

The formal definitions and names of these portfolio metrics are:

$$\text{Returns} = \mathbf{x}'_{\Pi} = \frac{\Delta \mathbf{x}_{\Pi}}{\Delta t} \quad (34)$$

Where t is selected to be some time period, by default a year. $\mathbf{x}'$ is a random variable as manuscripts will have some distribution of returns and the exact return for a given manuscript cannot be known a priori.

$$\text{Expected return} = \mu_{\Pi} = \mathbb{E}[\mathbf{x}'_{\Pi}] = \int_{-\infty}^{\infty} \mathbf{x}'_{\Pi} \cdot f_{\mathbf{x}'}(\mathbf{x}'_{\Pi}) d\mathbf{x}'_{\Pi} \quad (35)$$

Where $f_{\mathbf{x}'}(\mathbf{x}')$ is the probability density function of $\mathbf{x}'$, not known a priori and gathered from weighted citation (section 2.1) and capital graph (section 5.1) data.

6.6 Volatility and Risk Asymmetry

Given the definition for portfolio returns above, the volatility of returns can also be quantified for a portfolio or individual manuscript.

$$\text{Volatility} = \sigma_{\Pi} = \sqrt{\mathbb{E}[\mathbf{x}'_{\Pi} - \mu_{\Pi}]^2} \quad (36)$$

Volatility or standard deviation, measures how much on average returns differ from the mean return and is a useful risk measure in marketplaces and of portfolio holding entities.

In addition, risk asymmetry, or the asymmetry of academic capital returns can also be measured. This measure can reveal for academic fields, what the underlying skew of distribution of returns is, and whether outliers (fat tail effects) are common or rare.

$$\text{Risk Asymmetry} = \gamma_{\Pi} = \frac{\mathbb{E}[\mathbf{x}'_{\Pi} - \mu_{\Pi}]^3}{\sigma_{\Pi}^3} \quad (37)$$

6.7 Sharpe's Ratio and Price to Earnings Ratio

One useful quantity for valuable assets is the returns as a ratio to the volatility, also known as Sharpe's ratio. Sharpe [1966]

$$\text{Sharpe's ratio} = \frac{\mu_{\Pi}}{\sigma_{\Pi}} \quad (38)$$

In Liberata, because all academic capital is generated from manuscripts, and there is no other source of inflation or academic capital generation, the risk free return is 0, which leads to a simplified version Sharpe's ratio.

A natural analogy to the price to earnings ratio for capital assets in finance is the academic capital to returns ratio, which we invert to academic returns to capital ratio (ARC) for more intuitive interpretability.

$$\text{ARC} = \frac{\mu_{\Pi}}{\mathbf{x}_{\Pi}} \quad (39)$$

This measure gives a simple and intuitive way to tell if a manuscript is in a growth phase of impact (high ARC), tapering phase (low ARC), or stagnant (0 ARC).